ELEC-G001-A | Practice

Power distribution in a sensor circuit | 8 marks for the complete question

Figure 1 shows an ideal 18 V battery connected to a resistor network. All resistors are ohmic and their values remain constant. Show your working and include units. Use the same circuit for Questions 1–4.

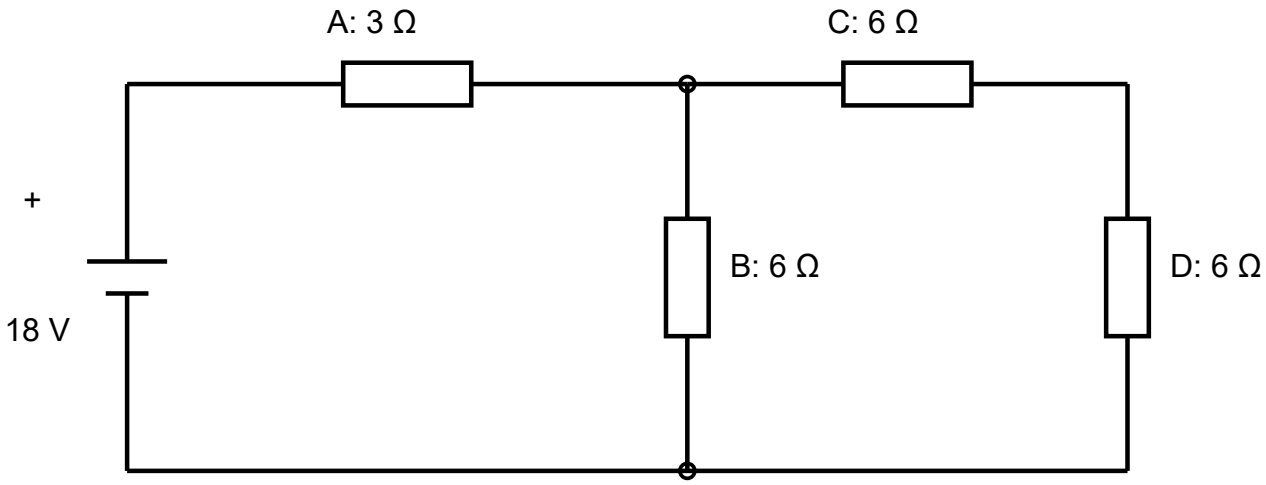


Figure 1

Question 1 (2 marks) Calculate the total resistance of the circuit.

Question 2 (2 marks) Calculate the current through resistor B.

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Continue the same question. Figure 1 is repeated for reference; use the same component values.

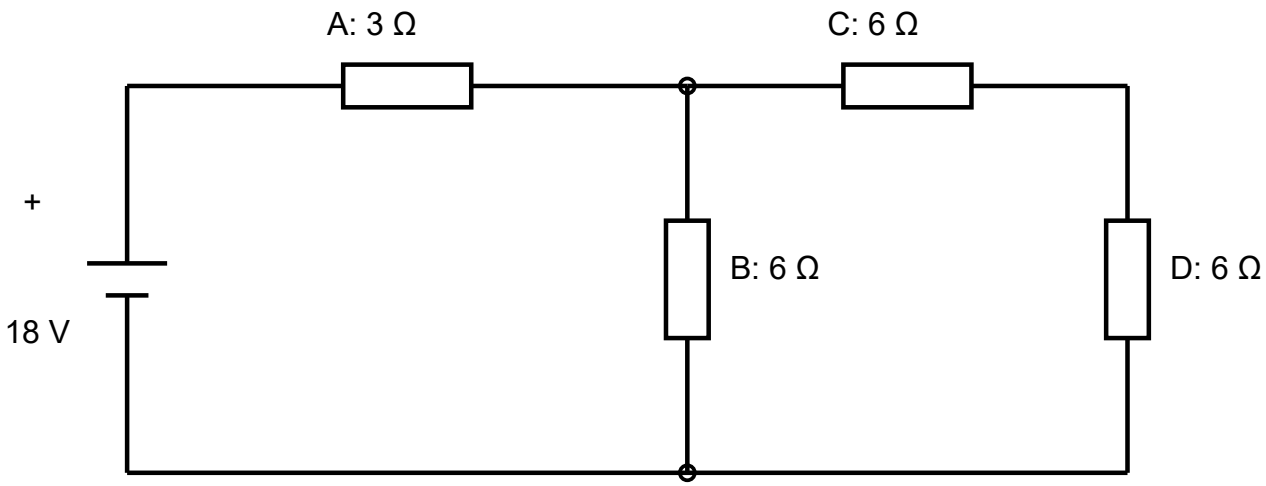


Figure 1

Question 3 (2 marks) Calculate the voltage across resistor A.

Question 4 (2 marks) Calculate the power dissipated in resistor D.

ELEC-G001-B | Practice

Opening one parallel branch | 10 marks for the complete question

Figure 1 shows an ideal 12 V supply. Switch S is initially closed and has negligible resistance. All resistor values remain constant. Use the closed-switch circuit for Questions 1–3. Only Question 4 changes the switch position.

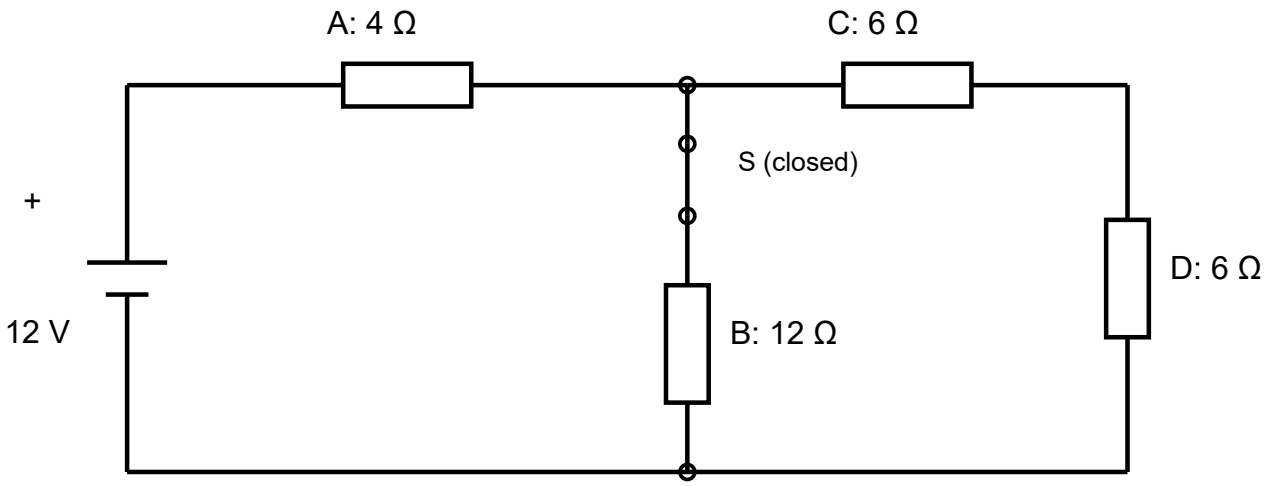


Figure 1

Question 1 (2 marks) Calculate the total resistance with S closed.

Question 2 (2 marks) Calculate the current through B with S closed.

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Continue the same question. Figure 1 is repeated for reference; use the same component values.

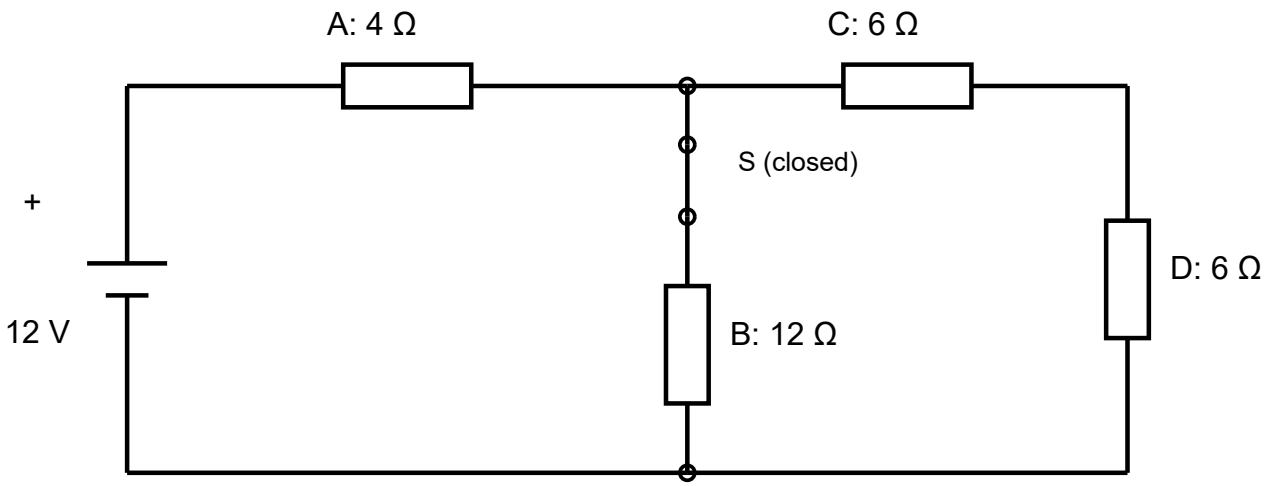


Figure 1

Question 3 (2 marks) Calculate the voltage across A with S closed.

Question 4 (4 marks) Calculate the power dissipated in D before and after S opens. Explain why the power in D can increase even though the battery current decreases.

ELEC-G002-B | Practice

Repair an incorrect divider | 4 marks for the complete question

A student uses an ideal 9.0 V battery and three identical 900 ohm resistors. Figure 1 shows the student's circuit. The target is 6.0 V across one resistor and 3.0 V across each of the other two.

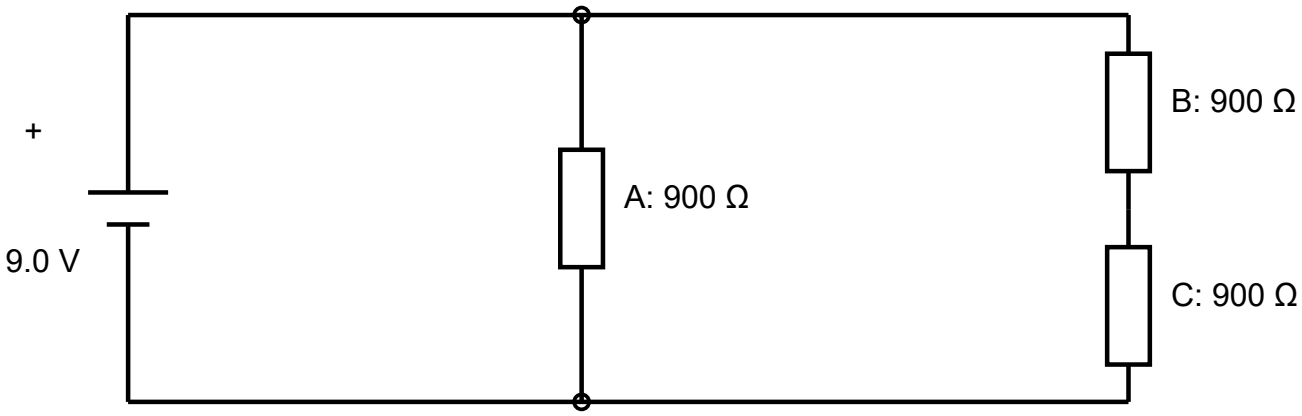


Figure 1

(a) (2 marks) Determine the actual voltage across each resistor in Figure 1.

(b) (2 marks) Draw a corrected circuit using all three resistors. Show that it meets the target.

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ELEC-G002-A | Practice

Design a two-to-one voltage split | 4 marks for the complete question

You have an ideal 6.0 V battery, three identical 100 ohm resistors and ideal connecting wires. Figure 1 provides the circuit symbols; it does not show a connection arrangement.



Use these circuit symbols. Connecting wires are ideal.

Figure 1

(a) (2 marks) Using all three resistors, draw a complete circuit so that one resistor has 4.0 V across it and each of the other two has 2.0 V. Label these voltages.

(b) (2 marks) Calculate the current supplied by the battery and use it to verify the required voltage split.

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